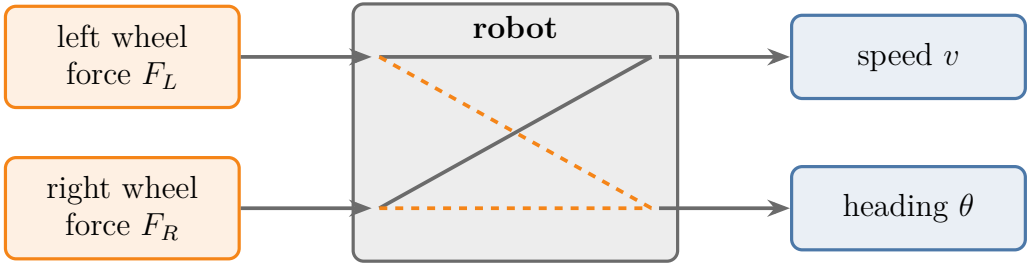
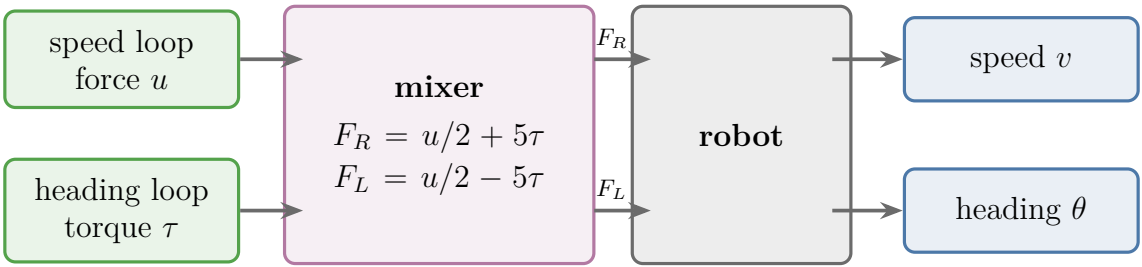
(a) two wheels, two outputs: one MIMO system



each wheel pushes the speed (solid) and turns the heading (dashed)

(b) a mixer in front: two SISO loops



u moves only the speed: $1/(8s + 2)$; τ turns only the heading: $10/s^2$